THE UNIVERSITY OF AZAD JAMMU AND KASHMIR
Department of Software Engineering

LAB REPORT 1

Use Case Diagrams

Modelling the online reservation system and credit card processing in UML

COURSE	Software Construction & Development (Lab)
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INSTRUCTOR	Engr. Dr. Asma Javaid
TOOL USED	Visual Paradigm Community Edition
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1. Introduction

This laboratory report documents the system modeling, actor identification, and use case specifications executed for Laboratory 01 of the Software Construction & Development course. The lab covers two software system domains: an Online Ticket Reservation System (Task 1) and a Point-of-Sale (POS) Credit Card Processing System (Task 2).

Table 1: Summary of lab tasks

Task	System	Interaction	Diagram
1	Online Ticket Reservation System	Web user interface	UML use case diagram
2	Credit Card Processing System	POS terminal hardware	UML use case diagram

2. Task 1: Online Ticket Reservation System

2.1 Overview and Actor Identification

- **Customer (Primary Actor):** Browses travel/event schedules, selects seats, and processes online payments.
- **System Admin (Secondary Actor):** Manages ticket cancellations and releases seat inventory.
- **Core Use Cases:** Search Ticket, Book Ticket, Process Payment, and Cancel Booking.

2.2 Solution Diagram

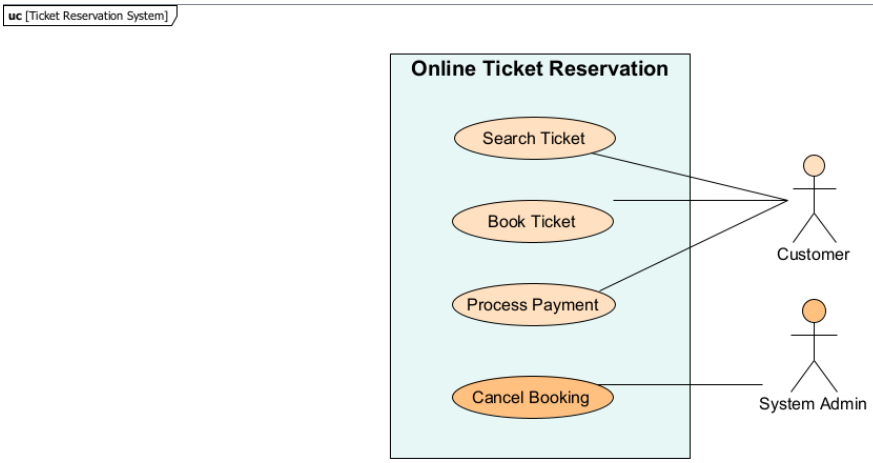


Figure 1: Use case diagram of the Online Ticket Reservation System

2.3 Key Use Case Specification

Table 2: UC-01 Book Ticket

Field	Details
Use Case Name	UC-01: Book Ticket
Primary Actor	Customer
Pre-Conditions	Customer has searched for tickets and selected a seat.
Main Flow	1. Customer selects ticket and clicks 'Book Ticket'. 2. System locks seat inventory for 10 minutes. 3. Customer confirms payment details.
Post-Conditions	Booking record status set to 'Pending Payment'.
Exceptions	Selected seat concurrently booked: system prompts re-selection.

3. Task 2: Credit Card Processing System (Case Study)

3.1 Overview and Actor Identification

- **Customer (Primary Actor):** Selects items, swipes credit card, and signs physical purchase receipt.
- **Vendor (Operational Actor):** Generates purchase bill and hands printed balance statements to customer.
- **Card Reader (Hardware Actor):** Reads card data, communicates with bank server, and prints statements.
- **Core Use Cases:** Select Item, Swap Card, Generate Bill, Process Amount Transaction, Issue Balance Statement, Sign Receipt.

3.2 Solution Diagram

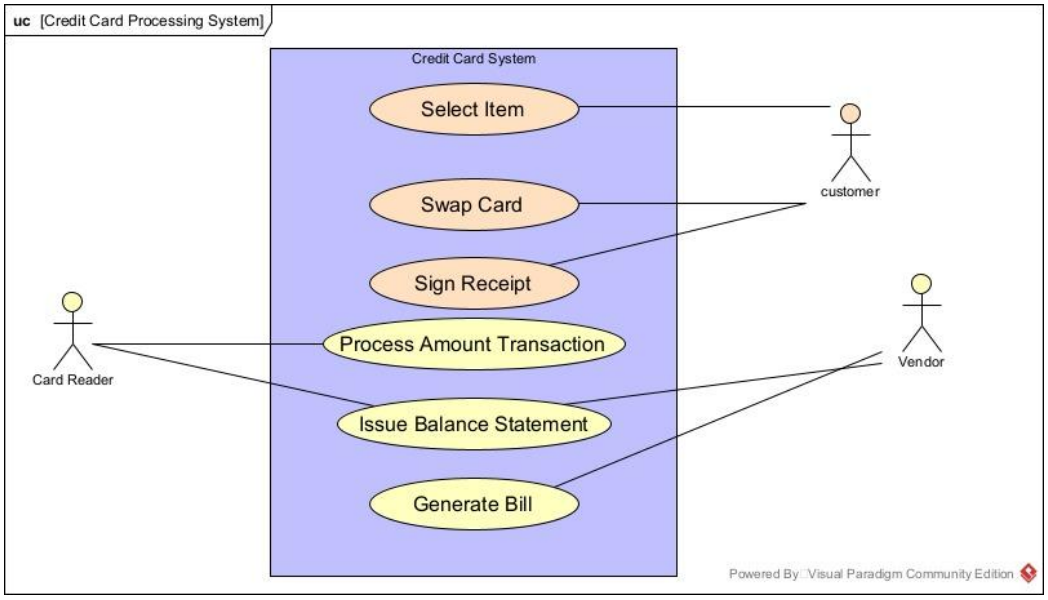


Figure 2: Use case diagram of the Credit Card Processing System

3.3 Key Use Case Specification

Table 3: UC-02 Process Amount Transaction

Field	Details
Use Case Name	UC-02: Process Amount Transaction
Primary Actors	Card Reader (Hardware), Vendor, Customer
Pre-Conditions	Card swiped in POS terminal; total bill calculated by Vendor.
Main Flow	1. Card reader reads card data and sends bank authorization. 2. Bank validates limit and approves transaction amount. 3. Balance decremented and confirmation returned.
Post-Conditions	Transaction completed and merchant account credited.
Exceptions	Insufficient balance: card reader displays 'Declined'.

4. Summary and Conclusion

- **System Differences:** Task 1 relies on web UI user interaction, whereas Task 2 operates through physical POS terminal hardware interaction.
- **Validation Model:** Ticket booking uses temporary seat holds (asynchronous), whereas credit card processing requires real-time synchronous bank verification.

In this lab, UML Use Case diagrams were analyzed and documented for online ticketing and POS transaction systems. System boundaries were established, human and hardware actors were categorized, and core functional flows were documented into concise specifications.